

have or plan to introduce. If most of the PCs in your office are going to be desktops then I suggest you go for wired networking. However, if most people are to use laptops, then wireless is definitely the answer—primarily for the convenience it offers.

When going wired, think long term—I would recommend going the Gigabit Ethernet way. It will significantly improve network speed and also offer higher bandwidth, thus taking care of congestion. When putting up the network, choose good quality products—don't try to cut costs and settle for anything because it's cheaper than the rest.

Switches are getting cheaper, and hence I encourage buying them over hubs. Vendors such as 3Com, D-Link, Netgear, and HP have good LAN Switches and an unmanaged 16 port Gigabit Ethernet switch won't cost more than Rs 16,000. Pair these Gigabit switches with proper Category 6 (CAT 6) cables to extract maximum benefits. CAT5E can also be used with Gigabit switches, but I still recommend opting for CAT 6 cables.

Deploying a wireless network requires less time than



The Linksys WRT54G is perfect for small wireless 802.11g networks



The D-Link DIR655 is a good 802.11n wireless router

wired network and there are few considerations you need to keep in mind.

To maximise the signal coverage area, wireless devices have to be installed high up. When planning seating, locate a spot such that the wireless device falls in the centre and the client devices (laptops or cell phones) fall within the signal coverage radius. Avoid installing the wireless device near wooden cabinets, metal objects or at desk level as this will lead to attenuation of signal and hence lesser coverage. Mounting the devices high results in signal reflecting off the ceiling thus improving signal coverage.

Most devices come with omni-directional antennas and broadcast signal around the router; good when you put the router in the centre of the room but not ideal when the device is mounted on a wall. You can buy a high-gain directional antenna which will direct maximum signal in the direction you want. There might be situations where some part of your office might not get enough signal, use repeaters to improve the strength. Network security is a concern with wireless networks—ensure your devices use the highest possible security measure possible. WPA2 offers good protection and is available with almost all new Wi-Fi devices.

Vendors such as Linksys, D-Link and ASUS have some real good products. Linksys has two ranges of products—one aimed at home users and other at professional; opt for the professional (WRT) series—the WRT-54G is quite a popular product. D-Link has come out with

UPS Types and their Characteristics							
	Practical Power Range (KVA)	Voltage Conditioning	Efficiency	Cost Per VA	Benefits	Limitations	Good For
Standby	0–0.5	Low	Very High	Low	Low Cost, High Efficiency and compact	Uses Battery during brownout, Impractical over 2kVA	Best Value for Personal Workstation
Line Interactive	0.5–5	Design Dependent	Very High	Medium	High Reliability, high Efficiency and good voltage conditioning	Impractical Over 5kVA	Popular UPS for high reliability, good for harsh power environments and for use with Servers
Standby Online Hybrid	0.5–5	High	Low	High	Excellent Voltage conditioning	Low efficiency, Low reliability, High cost and impractical over 5kVA	Line Interactive provides better reliability and similar power conditioning at better value
Standby Ferro	3–15	High	Low	High	Excellent Voltage conditioning and High reliability	Low efficiency, Unstable in combination with some loads and generators	Limited usage due to low efficiency and instability issues are a problem
Double Conversion Online	5–5000	High	Low	Medium	Excellent Voltage Conditioning and ease of paralleling	Low efficiency, Expensive under 5kVA application	Well suited for N+1 design
Delta Conversion	5–5000	High	Low	Medium	High Efficiency and Excellent Voltage conditioning	Impractical under 5kVA	High efficiency reduces the substantial life cycle cost in large installations

some real good products and they were awarded in our recently concluded test. The D-link DIR-655 has loads of features, is compatible with 802.11N devices and offers blistering performance.

Personal Computers

When it comes to buying a PC for your office, you can either go branded or get them assembled from your local assembler. Branded PC players do offer some benefits—tighter quality control, extensive after sales support, and if you bargain hard, good corporate discounts. Sadly, the configuration always falls short of performance. On the other hand, assembling your PC gives you the flexibility of building it to your specifications and requirements. Also, for the same price you can always build better performing machine than branded PCs. If you have a trusted knowledgeable assembler then I would definitely say go assembled else you may consider buying branded PCs.

What is an ideal Office PC configuration? There is no one configuration which can be termed as the ideal office PC configuration. The configuration will largely depend on the kind of work which will be done on it. Couple of years ago building a PC was quite simple; plunk a high end processor and add loads of memory to build a high performance machine. However, today this doesn't completely hold true. Most new applications incorporate visual effects and have intuitive interfaces—for example, Windows Vista, Microsoft Office 2007 and so on. Running such visually heavy application requires incredible amount of computation power which the processor can't handle. In other words, a graphics solution is important and can't be neglected.

A reasonably powerful, multi-purpose PC can be assembled within Rs 30,000. But going by general convention, you might be tempted to put in the best possible processor, around 2 GB of memory, an onboard graphics solution, large hard drive, CRT monitor and the regular stuff. Such a system would be inherently fast but will fail to deliver the goods when it encounters graphically taxing tasks. Here, I would suggest you go for reasonably good processor, buy any entry level graphics card from NVIDIA or ATI, put 2 GB of memory and from the money you saved by not buying the best processor buy an LCD monitor. Such a system would also deliver near good performance and importantly will be responsive under graphically taxing tasks.

Having a graphics card also has



An HP ProLiant ML100 series entry-level server for light to moderate loads—a good file and database server

PC Configurations Suitability To Task			
Applications Areas	Configuration No 1	Configuration No 2	Configuration No 3
Finance and Accounts work	Very Much	Yes, only if you run Financial Analysis, If so upgrade processor and remove graphics card	Not Recommended
Typical Office work (Internet browsing, Office suites)	Very Much	Not Recommended	Not Recommended
Data Entry, Call Centre	Very Much	Not Recommended	Not Recommended
Programming, DataBase	Can be used but not Recommended	Very much Recommended	Not Recommended
DTP, 2D design work (Photoshop, Corel, etc)	Not Recommended	Very much Recommended	Recommended only if 3D work is also done
3D modelling, Audio Video encoding, Media Creation	Not Recommended	Good for AV encoding	Very Much Recommended
Gaming parlours	Not Recommended	Can be used for older games	Very Much Recommended; change graphics card to Geforce series, 8800GT
Internet cafes	Very much Recommended	Not Recommended	Not Recommended

additional benefits. New generation graphic chips from NVIDIA and ATI are completely programmable and hence they can be made to do certain task that only CPUs could do. NVIDIA's CUDA and ATI's Brooke are the technologies are just emerging and hold true potential in days to come. For example, NVIDIA's CUDA is already available and cuts down computing-intensive tasks such as transcoding media from hours to just minutes. Even vendors such as Adobe have released applications which take advantage of this technology to cut down processing time and improve productivity.

Buy LCD displays—they are sleek and save a lot of desk space. They also require less power, and hence reduce electricity bills. Viewsonic, Samsung, LG, AOC and Dell have excellent products on offer.

I am listing three combination of critical components which determine their performance.

Configuration 1: AMD Athlon64 X2 5000+ AM2, Any AMD 780G or NVIDIA 8200 chipset motherboard from ASUS, Gigabyte or MSI, 512 MB DDR II RAM from Transcend or Kingston 160 GBhard drive space and 17" LCD display from Viewsonic, LG, Dell etc. This configuration along with common components should cost you more than Rs 20,500.

Configuration 2: Intel Core 2 Duo E7200 processor, NVIDIA 8200 or Intel P35 Chipset